

Letizia Girardi

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Research Interests

Longitudinal Multimodal Learning in Healthcare: multimodal deep-learning modeling integrating medical imaging with clinical data. Focus on generative AI models for temporal morphological analysis to quantify anatomical progression and improve diagnostic generalization.

Keywords: Multimodal Fusion, Longitudinal Modeling, Generative AI, Disease Progression Simulation, Disease Trajectory

Teaching Assistant

Autonomous System Agent (Master's Course) – Spring 2026 University of Trento
Provided hands-on support to students developing autonomous software agents using Python and LLM-integrated frameworks, including guidance on design, implementation, and debugging.

Calculus I (Bachelor's Course) – Winter 2024 University of Trento
Supported students through academic mentoring and problem-solving sessions for Analisi I, helping them master core concepts in calculus and mathematical analysis.

Education

University of Trento Trento, Italy
Ph.D. Student in Engineering Science Nov. 2025 – Present
Master's Degree in Computer Science (Bioinformatics) Sept. 2022 – Mar. 2025
Thesis: *Automatic Analysis of Perivascular Spaces in Parkinsonian Patients from Brain MRI Images*
Bachelor's Degree in Computer Science Sept. 2019 – Sept. 2022
Thesis: *Detection and Localization of Deep Video Inpainting*
Charles University - Prague, Czech Republic Sept. 2023 – Feb. 2024
Exchange Semester

Publications

- [3] **UNETR-RF: Reliability-guided multimodal MRI fusion for efficient brain tumor segmentation under domain shift**
Authors: Hafiza Ayesha Hoor Chaudhry, Selene Tomassini, **Letizia Girardi**, Luca Zigiottto, Silvio Sarubbo, Paolo Giorgini
Computerized Medical Imaging and Graphics - under peer review
- [2] **Deep segmentation of brain perivascular spaces in Parkinson's disease: informing decision support from clinical low-field MRI**
Authors: Selene Tomassini*, **Letizia Girardi***, Raffaella Di Giacobbe, Carlo Cosimo Quattrocchi and Paolo Giorgini
BMC Medical Informatics and Decision Making - under peer review
- [1] **A Case Study for the Automatic Supervision of Body-Weight Exercises: The Squat**
IEEE International Workshop on Sport, Technology and Research (STAR), Sept. 2023
Authors: Paolo Aliprandi*, **Letizia Girardi***, et al.
[DOI:10.1109/STAR58331.2023.10302651](https://doi.org/10.1109/STAR58331.2023.10302651)

**Equal contribution*

Awards & Honors

Selected Participant – HealthTech Week ETH Zürich

Aug. 2026 (upcoming)

Selected through a competitive application process to attend the intensive, hands-on week where students from Life Sciences/Medicine, Engineering, and Business/Economics collaborate to turn real clinical and industry problems into validated solutions, prototypes, and investor-ready pitches.

Selected Participant – Oxford Summer School on MLx Representation Learning & Gen AI *Jul. 2026 (upcoming)*

Selected through a competitive application process to attend the summer school at the University of Oxford.

Winner – Information Engineering Award - Collegio Ingegneri Venezia

Feb. 2026

Awarded for my Master's thesis as part of the competition promoted by the Collegio Ingegneri Venezia, recognizing outstanding theses in engineering focused on research, innovation, and sustainability. Selected as the winner for the Information Engineering macro-sector by the Committee on 25 February 2026.

20th Bernardo Nobile Award, Area Science Park – Trieste, Italy

Nov. 2025

Awarded an honorable mention for the master's thesis "Automatic measurement and analysis of perivascular spaces in parkinsonian subjects from brain MRI images."

Conferences & Workshops

Engineers Excellence Conference Venice

18 Apr. 2026

EXCELLENCE ENGINEERS CONFERENCE – 7TH EDITION – GIULIA CECCHETTIN AWARD

Presentation of thesis awarded as winner in the Information Engineering macro-sector.

Conversational AI in Health at IWSDS26

1 Mar. 2026

International Workshop on Spoken Dialogue Systems Technology

MedicinAI – Cycle of Seminars

- *Seminar I: AI and Clinical Data – From Fragmentation to a Unified Vision* *17 Oct. 2025*
- *Seminar II: AI Supporting Physicians – Diagnosis, Therapy, and Personalized Care* *6 Nov. 2025*
- *Seminar III: Reliability and Transparency – Regulating the Use of AI in Healthcare* *25 Nov. 2025*

MedicinAI - Workshop

4 Dec. 2025

AI and Decision Support in Healthcare: From Data to Clinical Impact

General co-chair & organizer: *Coordinated scientific sessions, speaker interactions, and interdisciplinary discussions involving AI and healthcare professionals.*

Data4SmartHealth 2025 – University of Bolzano

27 Nov. 2025

Focus on nuclear medicine and radiomics, with sessions on AI-driven image analysis and radiomics workflows.

Trieste Next – Science Research Festival

Sept. 2023

Selected for the Academy Scholarship; attended sessions on AI, education, and rehabilitation.

Connect Brain Vol. III, APSS Trento

Jun. 2024

Workshop on advanced neuroimaging and neurosurgical techniques.

Research Experience

Automated Summarization of First-Radiology Visit Reports

Python · PyTorch · Transformers · Hugging Face · NLP · Clinical Text

- Built an LLM pipeline to summarize radiology reports, extracting key findings with 92% recall on critical lesions and atrophy markers.
- Optimized prompt engineering and domain adaptation, reducing clinical hallucinations by 35% compared to baseline GPT-4 models.
- Enabled structured follow-up, reducing interdisciplinary communication time for clinicians by an average of 4 minutes per report.

Synthetic 3D CT Library for EVAR – Generative AI for Medical Imaging

Diffusion Models · Transformers · 3D CT Imaging · PyTorch · Python · ANTs/FSL

- Top-ranked participant (Top 5%) in MediTech 2026 – Vascular Surgery Challenge.
- Designed a generative AI framework to synthesize 150+ high-fidelity 3D CT volumes of pre-/post-EVAR anatomy.
- Developed diffusion-based pipelines achieving a Multi-Scale Structural Similarity Index (MS-SSIM) of 0.88, significantly expanding training data for segmentation tasks.

Automatic Analysis of Perivascular Spaces in Parkinsonian Patients from Brain MRI

3D MRI · Deep Learning Segmentation · 3D Slicer · FSL/ANTs · PyTorch · Python

- Designed DL models for PVS segmentation, achieving a **Dice Similarity Coefficient** > **0.65**, a 12% improvement over prior state-of-the-art methods.
- Implemented 3D reconstruction and volumetric quantification for longitudinal analysis of 200+ patient scans.
- Validated results in real-patients, finding significant correlations ($p < 0.05$) between PVS volume and clinical progression scores.

Technical Skills

Languages: Python (advanced), C/C++, SQL, MATLAB, R, Bash

Machine Learning & NLP: PyTorch, TensorFlow, Keras, Scikit-learn, Hugging Face Transformers, LlamaIndex, LangChain, Autogen Studio, Ollama, CUDA, TensorRT

Medical Imaging & Bioinformatics: 3D Slicer, FSL / ANTs, Scikit-bio, BLAST / BLASTP, OpenCV

Data Analysis & Visualization: NumPy, Pandas, Matplotlib, Seaborn, Plotly, StatsModels

Tools & Platforms: Git / GitHub, Docker, VS Code, Jupyter Notebook, LaTeX, Azure, SSH, VirtualBox, NEURON

Volunteering

TEDx Curator and Volunteer Coordinated speaker scouting and pitch evaluation, supported speakers during content development, and contributed to the organization and successful delivery of TEDx events.